Sentinel Validation & Timing Report

Everything that was tested, and how accurately it ran on time.

Revision 1.0 | 2026-08-06 | Backend test suite (pytest), benchmark harness (backend/benchmarks), and live end-to-end measurements on 127.0.0.1:8000.

1. Scope and Methodology

This report documents the empirical validation of the Sentinel platform across three layers: (1) the automated backend test suite (98 tests across 12 modules), (2) the repeatable benchmark harness covering rule-based inference, concurrency, storage I/O, AI providers, and cryptography, and (3) live end-to-end latency measurements of the running deployment. All timings were captured on the development host with Ollama 0.x serving local models on 127.0.0.1:11434 and the FastAPI backend on 127.0.0.1:8000.

Latency figures use wall-clock time (time.perf_counter) around full request/response cycles, means of repeated runs where the harness samples multiple entries, and compressed-time simulation for the crisis halt protocol as documented in the harness source.

2. Automated Test Suite

Result: 98 passed in 71.37s (12 test modules, 2026-08-06).

2.1 Coverage

- auth flow (registration, weak-password rejection, login, token lifecycle)
- crisis policy (trigger, cooldown, acknowledgement, escalation freeze)
- crisis security (cross-account isolation, signed links)
- export data (journal summaries, clinical notes, patient data)
- followups (assignment, proof upload, grading)
- journal API (content validation, summaries, resummarize)
- model registry (provider selection, fallback)
- overview priorities (triage ordering)
- plain insights (rule-based insight extraction)
- ring sensor (pairing, device tokens, data ingestion)
- risk engine (risk scores, crisis threshold)
- sync events (offline journal/mood sync, event store)

The suite exercises authentication and session security, the journal-to-insight pipeline, crisis trigger and cooldown semantics, risk scoring, ring pairing and ingestion, exports, follow-up workflows, offline sync, and the model registry. No tests were skipped; all 98 passed.

3. Benchmark Suite - Raw Timing Data

The harness produced 47 logged runs; 46 passed and 1 were expected-skip failures (Groq cloud inference without an API key). The full logbook is committed at backend/benchmarks/logbook_benchmark.csv.

3.1 Discrepancy Detection (rule-based engine)

50 curated profiles spanning text-biometric mismatches, matches, and edge cases (empty text, minimal text, 2,000-char

repetition, zero biometrics, crisis-text-with-calm-physiology). The engine returned a verdict on every profile in 0.1 ms.

Run ID	Component	Concurrency	Input Size	Latency (ms)	P/F
#0001	Discrepancy Detection	1	50 profiles	0.1	PASS
#0002	Discrepancy #38	1	4 words	0.1	PASS
#0003	Discrepancy #45	1	4 words	0.1	PASS
#0004	Discrepancy #18	1	5 words	0.1	PASS

Aggregate accuracy: 96.0% (TP=21, FP=2, TN=27, FN=0; precision 91%, recall 100%).

3.2 Crisis Engine Concurrency

Concurrent crisis simulators at 1, 5, 10, and 25 parallel instances. No dropped threads at any load; stage-2 escalation never fired before acknowledgment (correct behavior).

Run ID	Component	Concurrency	Input Size	Latency (ms)	P/F
#0005	Crisis Engine	1	1 concurrent	101.9	PASS
#0006	Crisis Engine	5	5 concurrent	102.8	PASS
#0007	Crisis Engine	10	10 concurrent	104.0	PASS
#0008	Crisis Engine	25	25 concurrent	109.3	PASS

3.3 Halt Protocol (compressed time)

Acknowledgment injected at 15, 45, and 65 simulated seconds. Stage-2/stage-3 email flags fired exactly as expected for each ack point; acknowledgment halted escalation in every case.

Run ID	Component	Concurrency	Input Size	Latency (ms)	P/F
#0009	Halt Protocol	1	ack @ 15 sim-s	750.0	PASS
#0010	Halt Protocol	1	ack @ 45 sim-s	2250.0	PASS
#0011	Halt Protocol	1	ack @ 65 sim-s	3250.0	PASS

3.4 Storage I/O Scalability

JSON (plain and Fernet-encrypted) and SQLite stores benchmarked at 10, 50, 100, and 500 profiles, plus 10 concurrent writers. Read/write round-trips scale sub-linearly to 500 profiles.

Run ID	Component	Concurrency	Input Size	Latency (ms)	P/F
#0012	JSON Storage (unencrypted)	1	10 profiles | 2.7KB	23.8	PASS
#0013	JSON Storage (encrypted)	1	10 profiles | 3.5KB	65.8	PASS
#0014	SQLite Storage	1	10 profiles | 12.0KB	34.6	PASS
#0015	JSON Storage (unencrypted)	1	50 profiles | 13.5KB	27.8	PASS
#0016	JSON Storage (encrypted)	1	50 profiles | 17.3KB	47.8	PASS
#0017	SQLite Storage	1	50 profiles | 12.0KB	47.6	PASS
#0018	JSON Storage (unencrypted)	1	100 profiles | 27.0KB	34.1	PASS
#0019	JSON Storage (encrypted)	1	100 profiles | 34.5KB	53.1	PASS
#0020	SQLite Storage	1	100 profiles | 20.0KB	60.8	PASS
#0021	JSON Storage (unencrypted)	1	500 profiles | 135.0KB	132.1	PASS
#0022	JSON Storage (encrypted)	1	500 profiles | 172.3KB	129.5	PASS
#0023	SQLite Storage	1	500 profiles | 64.0KB	181.3	PASS

#0024	JSON Concurrent Writes	10	100 profiles (10 threads x 10 each)	48.1	PASS

3.5 AI Provider Benchmark

Mock baseline is a fixed 50 ms compute simulation. The Ollama (local Mistral) runs are real local inference: a cold 100-word call at 5,076 ms (model warm-up), then warm 250-word at 1,449 ms and 500-word at 1,746 ms. Groq is recorded as an expected skip (no API key on the test host).

Run ID	Component	Concurrency	Input Size	Latency (ms)	P/F
#0025	AI Summarizer	1	100w (40 entries)	50.5	PASS
#0026	AI Summarizer	1	250w (5 entries)	51.3	PASS
#0027	AI Summarizer	1	500w (5 entries)	51.5	PASS
#0028	AI Summarizer	1	100w	5076.3	PASS
#0029	AI Summarizer	1	250w	1448.5	PASS
#0030	AI Summarizer	1	500w	1746.1	PASS
#0031	AI Summarizer	1	N/A (no key)	0.0	FAIL

3.6 Security Benchmark

PBKDF2 key derivation scales from 4.2 ms (10k iterations) to 155.6 ms (600k iterations) - the operator ceremony default of 600k stays under 200 ms per derivation. Fernet encryption/decryption is sub-millisecond through 10 KB payloads. JWT encode/decode is sub-millisecond at typical claims sizes. A full derive+encrypt+decrypt round-trip at 600k iterations completes in 168 ms.

Run ID	Component	Concurrency	Input Size	Latency (ms)	P/F
#0032	PBKDF2 Key Derivation	1	10000 iterations	4.2	PASS
#0033	PBKDF2 Key Derivation	1	50000 iterations	12.5	PASS
#0034	PBKDF2 Key Derivation	1	100000 iterations	25.2	PASS
#0035	PBKDF2 Key Derivation	1	300000 iterations	76.8	PASS
#0036	PBKDF2 Key Derivation	1	600000 iterations	155.6	PASS
#0037	Fernet Crypto Overhead	1	100B to 228B	0.4	PASS
#0038	Fernet Crypto Overhead	1	1000B to 1420B	0.3	PASS
#0039	Fernet Crypto Overhead	1	10000B to 13432B	0.5	PASS
#0040	Fernet Crypto Overhead	1	100000B to 133432B	1.8	PASS
#0041	JWT Auth Handshake	1	64B claims payload	40.2	PASS
#0042	JWT Auth Handshake	1	256B claims payload	0.5	PASS
#0043	JWT Auth Handshake	1	1024B claims payload	0.5	PASS
#0044	Full Crypto Round-Trip (600K iter)	1	derive+enc+dec	168.1	PASS
#0045	Encrypted Payload Overhead	1	100B plaintext	0.4	PASS
#0046	Encrypted Payload Overhead	1	1000B plaintext	0.4	PASS
#0047	Encrypted Payload Overhead	1	10000B plaintext	0.5	PASS

4. Timing Accuracy Analysis

4.1 Where latency comes from

The platform has two latency regimes. Deterministic subsystems - discrepancy detection, crisis state, storage, and cryptography - complete in 0.1-200 ms. The single dominant latency is local LLM inference, which costs 1.4-1.7 s warm and ~5.1 s cold on commodity hardware. This asymmetry is deliberate: rule-based safety checks run instantly, while the AI summary layer is an asynchronous, cacheable enrichment.

4.2 End-to-end API latency (live deployment)

Endpoint	What it measures	Latency (ms)
POST /auth/login (psychologist)	password derivation + JWT issue	397.5
GET /psychologists/patients	assigned-patient read	7.2
GET /crisis/state	crisis state read	28.1
POST /discrepancy/check	rule-based discrepancy evaluation	35.3
GET /ring/devices	device registry read	41.2
POST /triage	full triage pipeline incl. local LLM	4711.6

REST reads and the rule-based discrepancy check respond in 7-41 ms. Login at ~397 ms is dominated by password key derivation (deliberate cost for offline-attack resistance). The full triage pipeline at ~4.7 s includes one local LLM call; the dashboard reads the cached summary after that, so interactive triage stays fast.

4.3 Accuracy vs. latency trade-off

The discrepancy engine - the safety-critical classifier - achieves 96% accuracy with zero false negatives across the golden-set profiles at 0.1 ms per evaluation. Because it is rule-based and deterministic, its accuracy and latency are reproducible across runs, hardware, and offline environments - the property that matters most for a crisis-adjacent subsystem.

5. Pass/Fail Accounting

46/47 benchmark runs passed. The 1 recorded failures are the Groq cloud benchmark, which fails only because GROQ_API_KEY is not set on the test host - cloud inference is an opt-in deployment configuration, not a regression. All 98 automated tests passed with zero skips.

6. Conclusion

The validation evidence shows a platform whose safety-critical paths are deterministic, fast, and reproducible: 96% discrepancy accuracy at 0.1 ms, no dropped crisis threads at 25x concurrency, correct stage firing across all halt-protocol points, and sub-millisecond cryptography. The only slow path is local LLM summarization, which is bounded, cacheable, and off the interactive critical path. The benchmark harness is committed to the repository and can be re-run with a single command to reproduce every figure in this report.

Reproduce with: cd backend && python -m benchmarks.runner --csv benchmarks/logbook_benchmark.csv | python -m pytest tests -q. Companion documents: Research Paper (docs/sentinel_paper.pdf), White Paper (sentinel_whitepaper.pdf).